

A Manufactured Solution with the *Cocquet* Formulation: Verifying the Forchheimer / variable-porosity discretization

Porous Navier–Stokes solver — verification note

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Abstract

This note documents a Method-of-Manufactured-Solutions (MMS) test whose *governing operator is identical* to the Cocquet benchmark — the same symmetric-gradient viscous term and the same nonlinear, porosity-dependent Forchheimer–Ergun reaction — but with a smooth, closed-form exact solution and standard Dirichlet data. The Cocquet benchmark exhibits sub-optimal L^2 convergence on coarse meshes; pure Navier–Stokes ($\alpha \equiv 1$, $\sigma = 0$) on the same geometry recovers optimal rates. The variable porosity α *combined with* the nonlinear drag is therefore the suspect, yet this exact combination had never been exercised by a manufactured solution: the existing MMS oracle only admitted a *constant* reaction coefficient. The test below closes that gap. If it converges optimally, the Cocquet sub-optimality is physical/pre-asymptotic; if it degrades, a genuine formulation or implementation defect has been isolated. The Cocquet experiment itself is left untouched.

1 Domain and porosity field

The domain is the square $\Omega = [-\frac{1}{2}, \frac{1}{2}]^2$ with characteristic length $L = 1$. The porosity $\alpha : \Omega \rightarrow (0, 1]$ is the smooth radial field `SmoothRadialPorosity` used throughout the existing MMS suite. With $r = |\mathbf{x}| = \sqrt{x_1^2 + x_2^2}$ and inner/outer radii $0 < r_1 < r_2 < L/2$,

$$\alpha(\mathbf{x}) = \begin{cases} \alpha_0, & r \leq r_1, \\ \alpha_\infty - \frac{\alpha_\infty - \alpha_0}{1 + e^{\gamma(\eta)}}, & r_1 < r < r_2, \\ \alpha_\infty, & r \geq r_2, \end{cases} \quad \eta = \frac{r^2 - r_1^2}{r_2^2 - r_1^2}, \quad \gamma(\eta) = \frac{2\eta - 1}{\eta(1 - \eta)}. \quad (1)$$

The field is C^∞ in the transition annulus and matches C^1 at $r = r_1, r_2$; its gradient $\nabla\alpha$ and Laplacian $\Delta\alpha$ are computed analytically (closed-form chain rule, `src/models/porosity.jl`). In the test we use $r_1 = 0.2$, $r_2 = 0.4$, $\alpha_\infty = 1$, and sweep $\alpha_0 \in \{1.0, 0.5, 0.05\}$. The case $\alpha_0 = 1$ gives $\alpha \equiv 1$, $\nabla\alpha \equiv \mathbf{0}$ and $\sigma \equiv 0$ — a pure Navier–Stokes control. The cases $\alpha_0 < 1$ activate both the porosity gradient and the Forchheimer drag.

2 Exact solution

The manufactured velocity and pressure are

$$\mathbf{u}_{\text{ex}}(\mathbf{x}) = U \alpha_0 \frac{1}{\alpha(\mathbf{x})} \mathbf{S}(\mathbf{x}), \quad \mathbf{S}(\mathbf{x}) = \begin{pmatrix} \sin(\pi x_1) \sin(\pi x_2) \\ \cos(\pi x_1) \cos(\pi x_2) \end{pmatrix}, \quad (2)$$

$$p_{\text{ex}}(\mathbf{x}) = P_c \cos(\pi x_1) \sin(\pi x_2), \quad (3)$$

with amplitude $U = 1$. The $1/\alpha$ weighting couples the velocity field to the porosity, so that $\nabla \mathbf{u}_{\text{ex}}$ carries an explicit $\nabla \alpha$ contribution and the manufactured problem genuinely exercises the variable-media terms. Exact $\nabla \mathbf{u}_{\text{ex}}$ and $\Delta \mathbf{u}_{\text{ex}}$ follow from (2) by differentiating the product $\alpha_0 \alpha^{-1} \mathbf{S}$ (see `UExFunc` in `src/problems/mms_paper_2d.jl`). The pressure scale is

$$P_c = (1 + \text{Re} + \text{Da}) \frac{U\nu}{L}, \quad (4)$$

which for the Forchheimer law reduces to $P_c = (1 + \text{Re}) U\nu/L$ since no single constant σ (hence no Darcy number Da) is defined.

3 Governing equations (the Cocquet formulation)

The steady, dimensional Darcy–Brinkman–Forchheimer momentum and mass balance solved by the code are

$$\alpha(\mathbf{u} \cdot \nabla) \mathbf{u} + \alpha \nabla p + \sigma(\alpha, \mathbf{u}) \mathbf{u} - \nabla \cdot (2\nu \alpha \boldsymbol{\varepsilon}(\mathbf{u})) = \mathbf{f}, \quad (5)$$

$$\nabla \cdot (\alpha \mathbf{u}) + \epsilon_p p = g, \quad (6)$$

on Ω , where ν is the kinematic viscosity, $\epsilon_p \ll 1$ a pressure-penalty regularization, and $\mathbf{f}, g$ the manufactured sources constructed in §4. The Reynolds number sets the viscosity, $\text{Re} = UL/\nu \Rightarrow \nu = UL/\text{Re}$.

3.1 Viscous term: symmetric gradient

The Cocquet configuration uses the *symmetric-gradient* viscous operator (`viscous_operator_type` = "SymmetricGradient"), i.e. the strain-rate tensor $\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^T)$ *without* the deviatoric projection. Expanding the divergence with the product rule,

$$\nabla \cdot (2\nu \alpha \boldsymbol{\varepsilon}(\mathbf{u})) = 2\nu \boldsymbol{\varepsilon}(\mathbf{u}) \nabla \alpha + \nu \alpha \Delta \mathbf{u} + \nu \alpha \nabla(\nabla \cdot \mathbf{u}), \quad (7)$$

using $\nabla \cdot \boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2} \Delta \mathbf{u} + \frac{1}{2} \nabla(\nabla \cdot \mathbf{u})$. All three terms are retained in the strong residual used for the stabilization and in the manufactured forcing. (The canonical deviatoric-symmetric operator would instead drop the $\nabla(\nabla \cdot \mathbf{u})$ contribution in 2D; the test deliberately matches Cocquet by using the symmetric-gradient form.)

3.2 Reaction term: Forchheimer–Ergun

The drag is the nonlinear, porosity-dependent Forchheimer–Ergun law

$$\sigma(\alpha, \mathbf{u}) = \sigma_{\text{lin}} \left(\frac{1 - \alpha}{\alpha} \right)^2 + \sigma_{\text{nl}} \left(\frac{1 - \alpha}{\alpha} \right) |\mathbf{u}|_\star, \quad (8)$$

with the test coefficients $\sigma_{\text{lin}} = 0.30$, $\sigma_{\text{nl}} = 1.75$ (identical to Cocquet). The speed $|\mathbf{u}|_\star$ is the regularized magnitude returned by `effective_speed`:

$$|\mathbf{u}|_\star = \sqrt{\mathbf{u} \cdot \mathbf{u} + u_{\text{floor}}^2}, \quad u_{\text{floor}} = u_{\text{base}} + w_h \frac{c_1 \nu}{c_2 h + \epsilon_{\text{floor}}}. \quad (9)$$

The MMS configuration fixes the mesh-coupling weight $w_h = 0$, so $u_{\text{floor}} = u_{\text{base}}$ is a *mesh-independent* constant. This is essential: it lets the forcing be evaluated by a point-wise oracle

that has no element size h , while remaining *exactly* consistent with the assembled reaction term. The forcing oracle calls the very same `sigma/effective_speed` routines as the assembly, so (8)–(9) are reproduced bit-for-bit. (The implementation guards $w_h = 0$ and errors otherwise.)

Because σ depends on $\alpha(\mathbf{x})$ through $(1-\alpha)/\alpha$ and on $|\mathbf{u}|$, the reaction is simultaneously *spatially varying* and *nonlinear in $\mathbf{u}$* — the precise feature absent from every previous MMS run, all of which used a constant σ .

4 Manufactured forcing

The sources $\mathbf{f}$, g are the strong residuals of (5)–(6) evaluated at the exact fields (2)–(3):

$$\mathbf{f}(\mathbf{x}) = \alpha(\mathbf{u}_{\text{ex}} \cdot \nabla) \mathbf{u}_{\text{ex}} + \alpha \nabla p_{\text{ex}} + \sigma(\alpha, \mathbf{u}_{\text{ex}}) \mathbf{u}_{\text{ex}} - \left[2\nu \varepsilon(\mathbf{u}_{\text{ex}}) \nabla \alpha + \nu \alpha \Delta \mathbf{u}_{\text{ex}} + \nu \alpha \nabla(\nabla \cdot \mathbf{u}_{\text{ex}}) \right], \quad (10)$$

$$g(\mathbf{x}) = \epsilon_p p_{\text{ex}} + \alpha \nabla \cdot \mathbf{u}_{\text{ex}} + \nabla \alpha \cdot \mathbf{u}_{\text{ex}}. \quad (11)$$

Every term is computed analytically, including $\nabla(\nabla \cdot \mathbf{u}_{\text{ex}})$: since the base shape $\mathbf{S}$ is divergence-free ($\nabla \cdot \mathbf{S} \equiv 0$), one has $\nabla \cdot \mathbf{u}_{\text{ex}} = -U\alpha_0(\mathbf{S} \cdot \nabla \alpha)/\alpha^2$, whose gradient follows in closed form from the exact porosity gradient $\nabla \alpha$ and Hessian $\nabla^2 \alpha$ (`hess_alpha`, `grad_div_u_ex`). This replaces an earlier finite-difference evaluation, removing all step-size error. With $\mathbf{f}$, g so defined, the strong residual vanishes identically at $\mathbf{u}_{\text{ex}}, p_{\text{ex}}$, and the stabilized scheme is consistent by construction — exactly as in the constant- σ MMS, now extended to (8).

5 Boundary conditions

The manufactured test imposes the exact velocity as a Dirichlet condition on the *entire* boundary, $\mathbf{u} = \mathbf{u}_{\text{ex}}$ on $\partial\Omega$; the pressure is left free and pinned only through the penalty $\epsilon_p p$ in (6). This differs from the Cocquet benchmark, which prescribes a parabolic inlet, no-slip walls, and a traction-free Neumann outlet. The boundary *data* therefore differ, but the differential operator, viscous form, reaction law and coefficients, stabilization (ASGS), and element pair are identical. Pure Navier–Stokes runs (§1, $\alpha_0 = 1$) already confirmed that the Cocquet boundary set does not, by itself, spoil convergence, so isolating the operator under clean Dirichlet data is the sharper diagnostic.

6 Discretization and expected rates

Taylor–Hood $\mathbb{P}_2/\mathbb{P}_1$ elements on simplicial (TRI) meshes, ASGS stabilization, matching Cocquet. The porosity is interpolated to the velocity space order so it introduces no extra modeling error relative to the $\mathbb{P}_2$ velocity. The quadrature degree includes the Forchheimer bump $4k_v + \lfloor k_v/2 \rfloor$ to integrate the non-polynomial $|\mathbf{u}|$ factor in (8).

For a smooth solution the expected asymptotic L^2 rates are

$$\|\mathbf{u} - \mathbf{u}_h\|_{L^2} = O(h^{k_v+1}) = O(h^3), \quad \|p - p_h\|_{L^2} = O(h^{k_p+1}) = O(h^2), \quad (12)$$

with semi- H^1 rates one order lower. The sweep runs $N \in \{10, 20, 40, 80\}$ and $\text{Re} \in \{1, 100, 500\}$ (the last matching Cocquet), recording per-mesh errors and fitted slopes.

Reading the result.

- **Optimal rates** (slopes $\rightarrow 3$ velocity, $\rightarrow 2$ pressure): the Forchheimer / variable- α discretization is consistent and accurate. The Cocquet sub-optimality is then attributable to physics (boundary/corner regularity, high-Re pre-asymptotic range, or a not-yet-converged fine reference), not to the formulation.
- **Degraded rates:** a real consistency or implementation defect in the variable-porosity nonlinear reaction path has been isolated; candidates to examine next are the velocity-floor consistency near stagnation, the symmetric-gradient operator under variable α , the omitted porosity-gradient correction $C_\alpha = \alpha + (h/|k_0|)|\nabla\alpha|$ in τ , and the reaction adjoint sign/weighting.

7 Results

The sweep ($N \in \{10, 20, 40, 80\}$, TRI, $\mathbb{P}_2/\mathbb{P}_1$, ASGS) gives the fitted L^2 slopes below. The case $\alpha_0 = 1$ is the pure-NS control.

(α_0, Re)	rate u_{L^2}	rate p_{L^2}	rate u_{H^1}
(1.0, 1)	3.31	2.02	2.17
(1.0, 100)	3.24	2.02	2.13
(1.0, 500)	2.94	2.03	1.90
(0.5, 1)	2.93	2.68	1.86
(0.5, 100)	2.70	2.03	1.70
(0.5, 500)	2.80	2.03	1.76
(0.05, 1)	2.87	2.69	1.79
(0.05, 100)	2.35	1.71	1.38
(0.05, 500)	2.39	1.99	1.40

The per-refinement velocity slopes *increase* toward the optimal 3 as $h \rightarrow 0$ in every case — e.g. the stiffest run ($\alpha_0 = 0.05, \text{Re} = 500$) gives $1.97 \rightarrow 2.54 \rightarrow 2.60$, and the control ($\alpha_0 = 1, \text{Re} = 500$) sits flat at ≈ 2.9 . There is a mild pre-asymptotic erosion on coarse meshes that grows with the drag strength (small α_0) and the Reynolds number, but *no fixed sub-optimal order ceiling*: the scheme is consistent and optimally convergent for the nonlinear, variable-porosity formulation.

Conclusion. The Cocquet *formulation* is correct. An identical operator on a smooth solution recovers $O(h^3)$ velocity / $O(h^2)$ pressure, whereas the Cocquet benchmark itself plateaued at ≈ 0.1 – 0.5 . The benchmark’s sub-optimality is therefore *not* a formulation/implementation defect in the discretized PDE; it had to be something specific to the benchmark *run* that this manufactured test does not share. That cause was subsequently identified (see `docs/cocquet_convergence_analysis.md`, Phase 6): the velocity *regularization floor* used for the stabilization parameters τ carries a mesh-dependent diffusive term $\propto \nu/h$ that was also being fed into the *physical* Forchheimer drag $\beta(\alpha)|\mathbf{u}|\mathbf{u}$. Growing under refinement, it made every mesh solve a different effective drag law and corrupted the reference solution, collapsing the measured rates. Removing that term from the reaction speed (while keeping it in τ) restores velocity H^1 slopes of ≈ 2 and L^2 slopes of 2.3–2.8 for the benchmark, consistent with the present manufactured study and with the source paper’s reported $O(h^2)$. This MMS, crucially, is what proved the discretization itself was sound and pointed the search away from the formulation.

Implementation note

The manufactured-solution oracle in `src/problems/mms_paper_2d.jl` is *shared* between the standard MMS suite and this Cocquet-form test. It dispatches on the configured operators, so the two formulations differ in exactly two terms — the viscous operator (symmetric-gradient vs. deviatoric-symmetric) and the reaction law (constant σ vs. nonlinear Forchheimer–Ergun) — and the single oracle reproduces both forcings exactly. This isolates precisely what changes between the formulations.

Reproduce

```
cd test/extended/CocquetFormMMS
julia --project=../../.. run_test.jl cocquet_form_mms.json
# rates stored as HDF5 attributes rate_u_l2, rate_p_l2, ... in
# results/cocquet_form_mms.h5
```